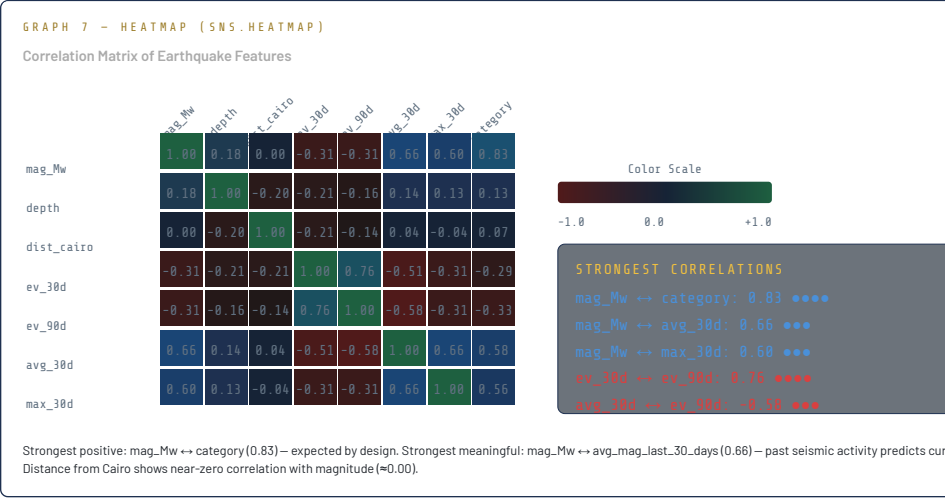
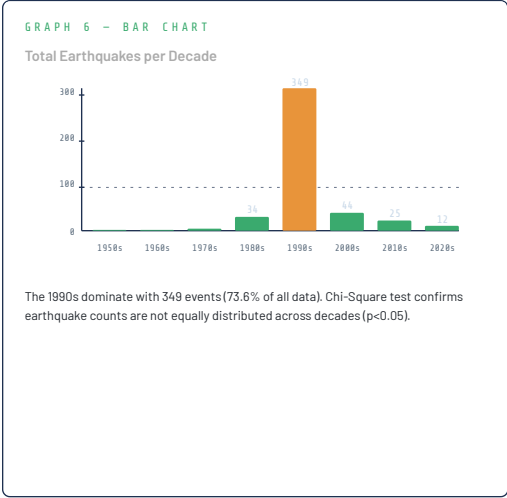
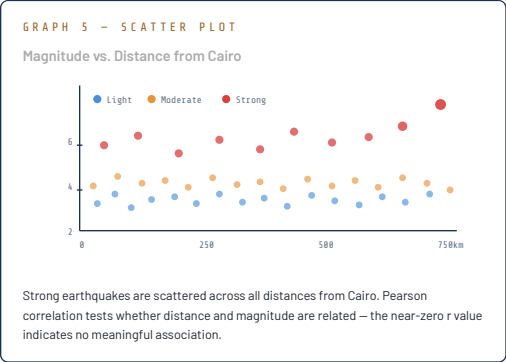
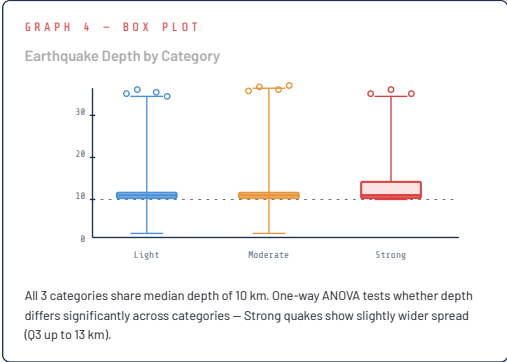
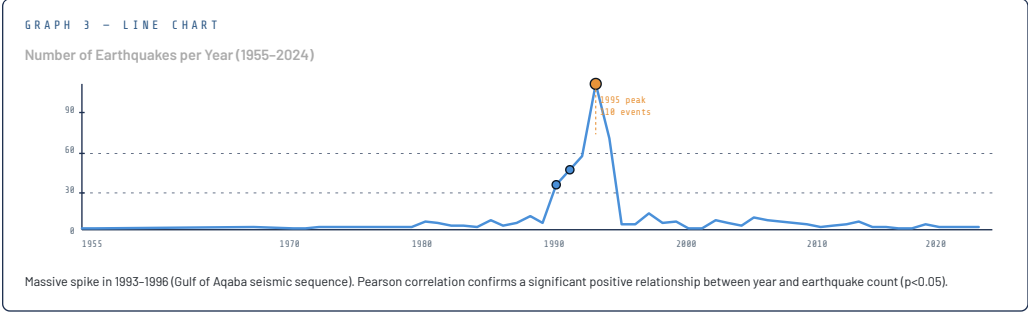
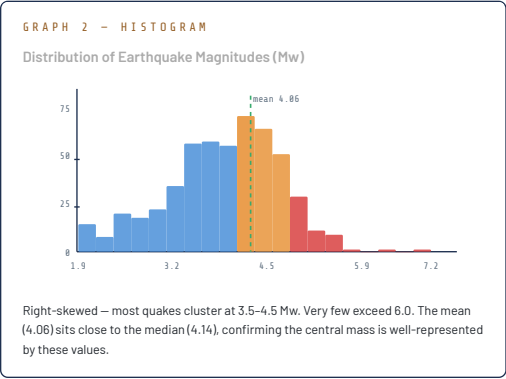
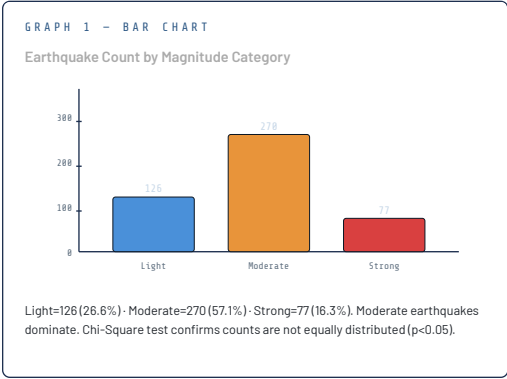




HYPOTHESIS TESTING

CHI-SQUARE · GRAPH 1 & GRAPH 6

Are earthquake counts equally distributed across magnitude and across decades?

H_0 : Counts are equal across all categories / all decades
 H_1 : Counts differ significantly — some categories or decades dominate

$P < 0.05 \rightarrow \text{REJECT } H_0$

ONE-WAY ANOVA · GRAPH 4

Does earthquake depth differ across Light, Moderate, and Strong categories?

H_0 : Average depth is the same for all three magnitude categories
 H_1 : At least one category has a significantly different average depth

$P < 0.05 \rightarrow \text{REJECT } H_0$

PEARSON CORRELATION · GRAPH 3 & GRAPH 5

Does earthquake frequency increase over time, and does distance from Cairo correlate with magnitude?

H_0 : No relationship between year & count; no relationship between magnitude and distance
 H_1 : Statistically significant linear relationships exist in both cases

$P < 0.05 \rightarrow \text{REJECT } H_0$

DESCRIPTIVE STATISTICS

COLUMN	MEAN	MEDIAN
Magnitude (Mw)	4.059	4.140
Depth (km)	11.138	10.000
Distance (km)	349.710	367.057

KEY INSIGHT

81% of all earthquakes occurred at exactly 10 km depth — a sensitive value indicating the Red Sea's characteristic shallow rift tectonic column shows near-zero variance across all 3 categories.

TOP CORRELATIONS WITH MAGNITUDE

$\text{mag_category_3classes}$
 $\text{avg_mag_last_30_days}$
 $\text{max_mag_last_30_days}$
 $\text{events_last_30_days}$

KEY FINDINGS

73.6%

1990s share

349 of 473 events occurred in the 1990s — driven by the intense 1992–1996 Gulf of Aqaba seismic sequence centered on 1995 (110 events in one year)

81%

at 10km depth

383 of 473 earthquakes recorded at exactly 10 km — the sensor default value, confirming this region's shallow rift tectonics rather than deep subduction

0.83

top correlation

Magnitude and category have $r=0.83$ — the strongest correlation in the dataset. Past average magnitude (avg_30d) is the best independent predictor at $r=0.66$

3 tests

per-graph stats

Chi-Square (Graphs 1 & 6), Pearson correlation and one-way ANOVA (Graph 4) all returned $p < 0.05$ confirming unequal distributions, temporal differences